Desert Terrain Semantic Segmentation Using Synthetic Data

Duality AI Offroad Semantic Scene Segmentation Challenge

Team: AI LONE STARS

Final mIoU: 64.50%

"From 2.37% to 64.50% — A Multi-Phase Journey to Robust Off-Road Perception"

Challenge: Train a semantic segmentation model on synthetic desert environment data
to accurately classify 10 terrain classes in unseen environments.

1. Introduction

Autonomous navigation in off-road environments requires robust, pixel-level terrain classification. This project addresses the challenge of segmenting **10 distinct desert terrain classes**, including rare and small objects like Logs and Rocks. By training on high-fidelity synthetic data generated from Duality AI's Falcon simulation platform, we developed a model capable of generalizing to complex, unseen environments, providing a critical perception layer for autonomous vehicles navigating hazardous terrains.

The core challenge lies in **domain generalization**: the model trains on one synthetic desert environment but must perform accurately on a completely different desert environment at test time. Additionally, extreme class imbalance (Sky covers ~37% of pixels while Logs cover only ~0.3%) makes this a particularly demanding segmentation task.

2. Approach

We employed a **multi-stage progressive refinement strategy** using the **DeepLabV3+** architecture from the `segmentation-models-pytorch` library. Each phase built upon the previous one, systematically addressing different aspects of the problem.

Phase 1: Foundation Training — 52.75% mIoU

Hardware: Laptop RTX 4050 (6GB VRAM) | **Backbone:** ResNet-34 | **Input:** 384×384

- **Loss:** CrossEntropyLoss with inverse-frequency class weights (capped at 8.0×)
- **Augmentations:** HorizontalFlip, RandomBrightnessContrast, HueSaturationValue, GaussNoise
- **Key Decision:** Used only CE loss (no Dice) to prevent model collapse during early training
- Mixed Precision (AMP) enabled for memory efficiency; `num_workers=0` for Windows compatibility

Phase 2: Rare Class Boost — 56.62% mIoU (+3.87%)

Hardware: Laptop RTX 4050 | **Loaded:** Phase 1 best checkpoint

- **Loss:** Combined CE (0.5) + Dice Loss (0.5) — added only after stable baseline
- **WeightedRandomSampler:** Logs 4×, Rocks 3×, Ground Clutter 2.5×, Dry Bushes 2×
- **New Augmentation:** CoarseDropout for occlusion robustness
- **Result:** Logs +5.1%, Rocks +7.5% — biggest single-phase improvement

Phase 3: High-Resolution Fine-Tuning — 59.11% mIoU (+2.49%)

Hardware: Laptop RTX 4050 | **Input:** 384→512×512

- **Gradient Accumulation:** Steps=2, effective batch size=8
- **Scheduler:** CosineAnnealingWarmRestarts (T_0=10, T_mult=2)
- **4-Augmentation TTA:** Original + H-Flip + Scale 0.75× + Scale 1.25×
- **Result:** Improved small object detection, Logs +4.45%

Final Step: Cloud Scaling & Ultra TTA — 64.50% mIoU (+5.39%)

Hardware: Google Colab Tesla T4 (15GB VRAM) | **Backbone:** ResNet-34→ResNet-101 | **Input:** 640×640

- Retrained with deeper backbone and higher resolution on more powerful GPU
- Same loss function and augmentation strategy as Phase 2
- Colab Phase 1: 58.93% → Colab Phase 2: 63%+ mIoU
- **8-Augmentation Ultra TTA:** Original + H-Flip + V-Flip + Scale 0.5× + Scale 0.75× + Scale 1.25× + Scale 1.5× + H-Flip+Scale 1.25×
- **Final Validation mIoU: 64.50%**

Architecture Summary

Component	Details
Architecture	DeepLabV3+ (Atrous Spatial Pyramid Pooling + Decoder)
Final Backbone	ResNet-101 (ImageNet pretrained)
Input Resolution	640×640 (final), progressive from 384→512→640
Loss Function	0.5 × CrossEntropy + 0.5 × Dice Loss
Optimizer	AdamW (lr=5e-5 for fine-tuning phases)
Scheduler	CosineAnnealingWarmRestarts
Inference	8-Augmentation Test-Time Augmentation (TTA)

3. Challenges & Solutions

Challenge 1: Complete Model Collapse (mIoU = 2.37%)

In early training, the model predicted *everything* as "Sky" or "Landscape" — the two dominant classes. The validation mIoU was just 2.37%, indicating the model had completely collapsed.

Root Cause: Dice loss combined with extreme class weights caused gradient instability during the warmup period.

Solution:

- Removed Dice loss entirely for Phase 1 — used only CrossEntropyLoss
- Increased learning rate from 1e-4 to 1e-3 to escape the collapsed state
- Verified mask remapping indices were correct (raw pixel values → class indices)

Result: Model immediately began learning all classes. Phase 1 achieved 52.75% mIoU.

Challenge 2: Extreme Class Imbalance (123× ratio)

Sky covers ~37% of all pixels while Logs cover only ~0.3% — a 123× imbalance.

Solution — 3-Pronged Approach:

- **Inverse-frequency class weights** capped at 8.0× to prevent gradient explosion
- **WeightedRandomSampler** oversampling images containing rare classes (Logs 4×, Rocks 3×, Ground Clutter 2.5×, Dry Bushes 2×)
- **Combined CE + Dice Loss** added only after stable baseline to refine boundaries

Result: Logs improved from 3.34% → 53.91% IoU. Rocks from 6.95% → 47.89% IoU.

Challenge 3: Limited GPU Resources (6GB VRAM)

Development was constrained to a laptop NVIDIA RTX 4050 with only 6GB VRAM.

Solution:

- **Mixed Precision (AMP):** Reduced memory usage by ~40%
- **Gradient Accumulation:** Steps=2 for effective batch size of 8
- **Progressive Resolution:** Started at 384×384, scaled to 512×512 as model stabilized
- **Cloud Scaling:** Transferred to Google Colab T4 (15GB) for final phase with ResNet-101 at 640×640

Result: Hardware upgrade alone contributed +5.39% mIoU improvement.

4. Results & Performance Metrics

The model showed consistent growth across all metrics. Notably, the **Logs** class improved from **21.90% to 53.91% IoU** through our multi-phase optimization, contributing to a final mIoU of **64.50%**.

Training Progression

Stage	mIoU	Key Change	Gain
Phase 1 (Laptop)	52.75%	Base training, CE loss, class weights	—
Phase 2 (Laptop)	56.62%	Oversampling + Dice loss	+3.87%
Phase 3 (Laptop + TTA)	59.11%	512×512 + 4-Aug TTA	+2.49%
Final (Colab + Ultra TTA)	**64.50%**	**ResNet-101 + 640×640 + 8-Aug TTA**	**+5.39%**

Total Improvement: 2.37% → 64.50% mIoU (+62.13%)

Summary Metrics

Phase	mIoU	Best Class	Worst Class
Phase 1	52.75%	Sky (97.07%)	Logs (21.90%)
Phase 2	56.62%	Sky (97.57%)	Logs (27.00%)
Phase 3	59.11%	Sky (97.86%)	Logs (31.45%)
Final (Apex)	**64.50%**	**Sky (97.67%)**	**Ground Clutter (38.97%)**

Final Per-Class IoU Progression

Class	Phase 1	Phase 2	Phase 3	Final (Ultra TTA)	Gain
Trees	77.13%	80.11%	82.35%	**85.30%**	+8.17%
Lush Bushes	64.66%	66.34%	67.66%	**69.26%**	+4.60%
Dry Grass	63.79%	65.89%	66.97%	**69.32%**	+5.53%
Dry Bushes	35.40%	42.57%	42.93%	**48.83%**	+13.43%
Ground Clutter	31.56%	33.23%	35.43%	**38.97%**	+7.41%
Flowers	53.75%	58.41%	60.56%	**65.76%**	+12.01%
Logs	21.90%	27.00%	31.45%	**53.91%**	**+32.01%**
Rocks	29.85%	37.33%	39.37%	**47.89%**	**+18.04%**

Landscape	52.36%	57.77%	59.58%	**68.08%**	+15.72%
Sky	97.07%	97.57%	97.86%	**97.67%**	+0.60%
Mean IoU	**52.75%**	**56.62%**	**59.11%**	**64.50%**	**+11.75%**

Biggest Improvements (Phase 1 → Final)

Class	Phase 1	Final	Absolute Gain
🪵 Logs	21.90%	53.91%	**+32.01%**
🪨 Rocks	29.85%	47.89%	**+18.04%**
🏜️ Landscape	52.36%	68.08%	**+15.72%**
🌿 Dry Bushes	35.40%	48.83%	**+13.43%**

Confusion Analysis

- **Dry Bushes ↔ Dry Grass:** Similar texture and color in desert environments causes mutual confusion
- **Ground Clutter ↔ Landscape:** Both are ground-level classes with overlapping visual features
- **Logs ↔ Rocks:** Similar size and shape in small instances; improved significantly with oversampling
- **Lush Bushes ↔ Trees:** Minor confusion at boundaries where canopy meets bush

5. Key Insights

1. Oversampling is the Single Biggest Lever

Targeted oversampling via WeightedRandomSampler was the single biggest contributor to mIoU gains (+3.87% in Phase 2). For severely imbalanced datasets, oversampling rare classes is more effective than loss function engineering alone.

2. Loss Scheduling Prevents Collapse

Dice loss is highly sensitive to class imbalance during early training. Adding it only after the model's CE baseline stabilized (Phase 2) prevented divergence. This "loss scheduling" approach is critical for multi-class segmentation with extreme imbalance.

3. Resolution Matters for Small Objects

Moving from 384→512→640 resolution progressively improved small object detection. Logs improved by +22.46% from Phase 1 to Final, largely driven by higher resolution enabling the model to see fine details.

4. TTA is Essential for Competition

Multi-scale, multi-flip TTA consistently added +1.5-3% mIoU at zero training cost. The 8-augmentation Ultra TTA was particularly effective for scale-variant classes like Logs and Rocks.

5. Progressive Training > Training from Scratch

Each phase loaded the previous best checkpoint rather than training from scratch. This curriculum-style approach allowed the model to build on stable representations rather than re-learning from random initialization.

6. Real-World Impact

This project demonstrates that **high-fidelity synthetic data from digital twins is a viable and cost-effective alternative** to real-world data collection for autonomous navigation. The multi-phase refinement approach ensures that even rare objects — critical for obstacle

avoidance in autonomous driving — are accurately detected without sacrificing performance on common classes.

- **Safety-Critical Detection:** Logs (53.91% IoU) and Rocks (47.89% IoU) are obstacles that could damage or immobilize a UGV. Our model detects these with reasonable accuracy despite their rarity.
- **Cost Efficiency:** Training entirely on synthetic data eliminates the need for expensive real-world data collection in remote desert environments.
- **Scalability:** The multi-phase approach can be applied to any new environment by generating synthetic data from Duality AI's Falcon platform.
- **Edge Deployment:** The model runs at ~9 images/second on a laptop GPU, making it feasible for real-time deployment on autonomous vehicles.

7. Tech Stack

PyTorch | segmentation-models-pytorch | Albumentations | ttach | Mixed Precision (AMP) | Google Colab | NumPy | OpenCV | Matplotlib | scikit-learn

8. Conclusion & Future Work

Conclusion

Through systematic multi-phase training, we improved from a **completely collapsed model (2.37% mIoU)** to a robust segmentation system achieving **64.50% mIoU**. The journey involved solving critical challenges including model collapse, extreme class imbalance (123× ratio), hardware constraints (6GB VRAM), and domain generalization.

The key insight is that **class imbalance must be addressed progressively** — first stabilize with CE loss, then add Dice loss and oversampling. Hardware scaling (laptop → Colab) provided meaningful gains, and Test-Time Augmentation is a free accuracy boost at inference time.

Future Work

Improvement	Expected Impact
Stronger Backbones: EfficientNet-B7 or ConvNeXt-Large	+2-4% mIoU
Ensemble Methods: Combine multiple model predictions	+1-3% mIoU
Self-Training: Pseudo-labels on high-confidence test predictions	+1-2% mIoU
Domain Adaptation: Adversarial training for synthetic→real gap	+2-5% mIoU
Higher Resolution: 768×768 or 1024×1024 with A100 GPU	+1-3% mIoU
Post-Processing: CRF or morphological operations for boundaries	+0.5-1% mIoU

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